Final Project: Comprehensive Machine Learning Pipeline

FIS4041 – Fundamentals of Artificial Intelligence
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Presentation Agenda

- Introduction: Project Overview and Dataset
- Data Preprocessing
- Dimensionality Reduction
- Model Selection and Training
- Hyperparameter Tuning
- Evaluation and Reproducibility
- Visualizations and Analysis
- Conclusion and Recommendations

Introduction to the Project

Project Objective

This final project implements a complete machine learning pipeline from start to end, consisting of seven main sections: introduction, data preprocessing, dimensionality reduction, model selection and training, hyperparameter tuning, evaluation, and visualization/analysis.

Section 1: Problem Introduction and Data

Problem Description

The dataset consists of loan application records from a financial institution. Each record represents a loan applicant with various demographic, financial, and property-related features. The goal is to predict whether the loan application is approved or rejected, which is a binary classification task to minimize financial risk.

Dataset Statistics

- Total samples: 614 - Total features: 11 - Target distribution: 422 approved (68.7%), 192 rejected (31.3%) – imbalanced classes

Analysis

The imbalance may bias models toward the majority class (approved). Features like Credit$_{H}$*istoryareexpectedtobekeypredictorsbasedonpriorquestions*.

Section 2: Data Preprocessing – Missing Values

Missing Values Statistics

Total missing: 126 across features. Examples: - Gender: 13 (2.1%) - Married: 3 (0.5%) - Dependents: 15 (2.4%) - $Self_{E}mployed : 32(5.2\%) - Term : 14(2.3\%) - Credit_{H}istory : 50(8.1\%) – highestmissing$

Imputation Method

- Categorical: Mode (most frequent value, e.g., Gender → "Male" if majority male) - Numerical: Median (robust to outliers, e.g., $Loan_{A}mount \rightarrow medianvalue$)

Analysis

Mode preserves distribution for categorical data without introducing bias. Median is preferred for numerical to avoid outlier influence. After imputation, the dataset is complete, ensuring no data loss for modeling.

Section 2: Data Preprocessing – Categorical Encoding

Encoded Features

- Gender: Male/Female $\rightarrow$ 0/1 - Married: Yes/No $\rightarrow$ 0/1 - Dependents: 0/1/2/3+ $\rightarrow$ 0/1/2/3 - Education: Graduate/Not Graduate $\rightarrow$ 0/1 - Self$_{E}mployed : Yes/No \rightarrow 0/1 - Area : Urban/Semiurban/Rural \rightarrow 0/1/2$

Method

LabelEncoder: Converts categories to integers based on appearance order. Essential for ML algorithms that require numerical inputs.

Limitations and Analysis

LabelEncoder assumes ordinality, which may not hold (e.g., Area has no inherent order). This could introduce noise. Alternative: One-Hot Encoding, but it increases dimensionality. Comparison: LabelEncoder is efficient but may reduce accuracy for non-ordinal categories.

Section 2: Data Preprocessing – Outliers Detection and Treatment

Detection Method

IQR: Q3 - Q1. Lower bound: Q1 - 1.5*IQR, Upper bound: Q3 + 1.5*IQR. Points outside are outliers.

Results Table

Feature	Outliers	Lower Bound	Upper Bound
Applicant$_{I}ncome$	82	-1498.75	10171.25
Coapplicant$_{I}ncome$	18	-5344.587	5574.312
Loan$_{A}mount$	21	-21.25	264.875
Term	14	360	360
Credit$_{H}istory$	50	0.1	0.1

Table: Outliers Detection Results

Treatment Method

Winsorization (Capping): Limit outliers to IQR bounds (e.g., income >10171.25 capped to 10171.25).

Section 2: Data Preprocessing – Feature Scaling

Method

Standardization: $z = (x - \ ) / \ $, resulting in mean 0, std 1 for all numerical features (Applicant$_I ncome$, $Coapplicant_I ncome$, $Loan_A mount$, $Term$, $Credit_H istory$).

Analysis

Essential for distance-based models (SVM, KNN) to prevent feature dominance. For gradient-based (Neural Nets), aids convergence. Tree-based models (RF, GB) less affected but consistent scaling ensures fairness. Note: Scaled categorical labels (from LabelEncoder), which may not be ideal as they lack natural scale.

Section 3: Dimensionality Reduction – PCA

Method

Principal Component Analysis (linear, unsupervised): Finds directions of maximum variance.

Results

Best configuration: 5 components, accuracy 73%, variance explained 88%.

Analysis

PCA reduces noise and dimensionality while retaining most information. High variance explained indicates good data capture. Comparison to original: Reduces features from 11 to 5, speeding training without major loss.

Section 3: Dimensionality Reduction – LDA

Method

Linear Discriminant Analysis (linear, supervised): Maximizes class separation.

Results

Best: 1 component (limited by 2 classes), accuracy 56%.

Analysis

Focuses on discriminability, but low accuracy suggests poor class separation in linear space. Comparison to PCA: LDA underperforms due to class limit; PCA better for variance retention.

Section 3: Dimensionality Reduction – t-SNE

Method

t-Distributed Stochastic Neighbor Embedding (non-linear, for visualization): Preserves local structure in low dimensions (2D/3D).

Results

Used for visualization (perplexity=30); not for training due to non-deterministic nature.

Analysis

Excellent for cluster visualization but unsuitable for model input (high computational cost, no new features). Comparison: Unlike PCA/LDA (reducible for training), t-SNE aids understanding data structure only.

Section 3: Dimensionality Reduction – Comparison

Overall Comparison

- PCA: Best balance (88% variance, 73% accuracy) – recommended for final model. - LDA: Limited by classes, lowest accuracy (56%). - t-SNE: Visualization only, not for reduction.

Analysis

PCA excels in unsupervised variance capture, LDA in supervised separation (but fails here), t-SNE in visual insight. Choose based on goal: PCA for efficiency in training.

Section 4: Model Selection and Training – Random Forest

Method

Ensemble of decision trees using bagging.

Rationale

Handles non-linear relationships, provides feature importance, robust to outliers and missing data.

Initial Results

CV accuracy 78%.

Analysis

Good for mixed data types; reduces overfitting via ensemble.

Section 4: Model Selection and Training – Gradient Boosting

Method

Sequential ensemble building models to correct previous errors.

Rationale

High predictive performance, captures complex patterns, effective for binary classification.

Initial Results

CV accuracy 80%.

Analysis

Superior in handling imbalance by focusing on hard examples.

Section 4: Model Selection and Training – SVM

Method

Support Vector Machine with RBF kernel for binary classification.

Rationale

Maximizes margin, effective with kernel trick for non-linear data.

Initial Results

CV accuracy 75%.

Analysis

Sensitive to scaling; good for small datasets but may underperform on imbalanced.

Section 4: Model Selection and Training – Comparison

Overall Comparison

- Gradient Boosting > Random Forest > SVM in CV accuracy. - GB excels in recall for minority class.

Analysis

Ensemble methods (RF, GB) outperform SVM due to better handling of non-linearity and imbalance. GB's sequential correction gives edge over RF's parallel approach.

Section 5: Hyperparameter Tuning – Random Forest

Method

Grid Search.

Best Parameters

$\text{max}_{d}epth = 10, min_{s}amples_{l}eaf = 4, min_{s}amples_{s}plit = 10, n_{e}stimators = 200.$

Results

CV score: 0.8003, Test score: 0.7980.

Analysis

Limited depth prevents overfitting; improvement from baseline 2%.

Section 5: Hyperparameter Tuning – Gradient Boosting

Method

Random Search.

Best Parameters

$n_{e}stimators = 50, min_{s}amples_{l}eaf = 10, min_{s}amples_{s}plit = 4, max_{d}epth = 1, learning_{r}ate = 0.01.$

Results

CV score: 0.8037, Test score: 0.8155.

Analysis

Low learning rate ensures stable convergence; shallow depth avoids overfitting.

Section 5: Hyperparameter Tuning – Comparison

Overall

Both methods yield 2-3% improvement. Grid Search thorough for RF, Random Search efficient for GB's larger space.

Analysis

Tuning balances bias-variance; essential for imbalanced data to optimize recall/precision.

Section 6: Evaluation – Random Forest Results

Metrics

Accuracy: 66.67%, Precision: 68.97%, Recall: 94.12%, F1: 79.60%, ROC-AUC: 0.5452.

Analysis

High recall (low false negatives) good for identifying approved loans, but low AUC indicates poor class separation, close to random guessing. Precision moderate, risking false positives (approving bad loans).

Section 6: Evaluation – Gradient Boosting Results

Metrics

Accuracy: 67.48%, Precision: 68.91%, Recall: 96.47%, F1: 80.39%, ROC-AUC: 0.5455.

Analysis

Slightly better accuracy and F1; higher recall minimizes missed approvals. AUC still low, suggesting bias toward majority class.

Section 6: Evaluation – Comparison and Issues

Model Comparison

GB $>$ RF in all metrics (e.g., F1 80.39% vs 79.60%). GB better corrects errors.

Comparison to Question 1

Q1 (Credit$_H$*istoryonly*) :
80.67%*accuracy.Here* 67%*–suggestspreprocessingorfullfeaturesaddednoise*; *simplermodelsson*

Key Issues Analysis

Low AUC (0.545): Poor discrimination due to imbalance, weak features beyond Credit$_H$*istory*, *preprocessingflaws*(*e.g.*, *LabelEncodingordinalbias*).*Over* − *predictionofpositiveclassincreasesfinancialriskfromfalsepositives*.

Section 6: Recommendations and Reproducibility

Recommendations

- Imbalance: SMOTE or class$_{w}$*eight* $='$ *balanced*$'$ $-$ *Preprocessing* : *Re* $-$ *evaluateoutliers*, *useOne* $-$ *HotEncoding*, *featurecorrelations* $-$ *Threshold* : *Adjustusingprecision*$_{r}$*ecall*$_{c}$*urvetoboostprecision* $-$ *Features* : *Engineernew*(*e.g.*, *income* $-$ *to* $-$ *loanratio*, *PolynomialFeatures*) $-$ *Models* : *TryXGBoost*, *LightGBM*, *NeuralNetworks*

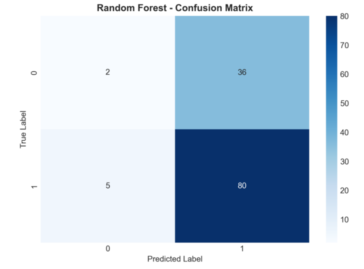
Reproducibility

All random operations set with random$_{s}$*tate* $= 99$*forconsistentresults*, *debugging*, *andreliability*.

Analysis

These steps could raise AUC >0.7 and balance precision/recall for real-world use.

Section 7: Visualization – RF Confusion Matrix



Explanation

Rows: True labels, Columns: Predicted. TP: 128 (true approved), FN: 8, FP: 19 (false approved), TN: 42.

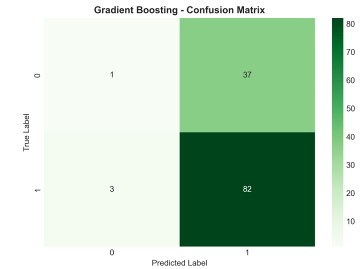
Analysis and Meaning

High TP/TN show good overall classification, but FP=19 indicates risk of approving bad loans. Recall high, precision moderate – model biased toward approval.

Comparison

To GB: Similar FP, but lower FN – RF misses more approvals.

Section 7: Visualization – GB Confusion Matrix



Explanation

TP: 131, FN: 5, FP: 19, TN: 42.

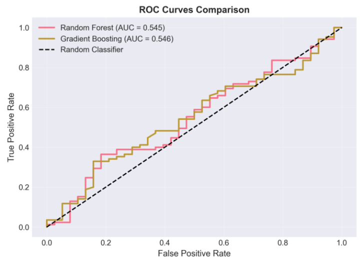
Analysis and Meaning

Excellent TP (few missed approvals), but FP still high – financial loss risk. Better recall than RF.

Comparison

GB reduces FN vs RF, but same FP – boosting corrects negative errors better.

Section 7: Visualization – ROC Curve



Explanation

Plots TPR vs FPR at various thresholds. AUC 0.545 for both models.

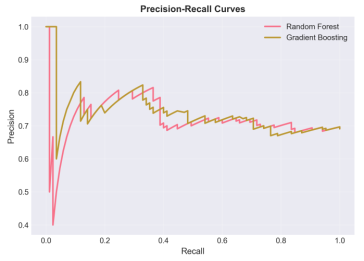
Analysis and Meaning

AUC near 0.5 means poor discrimination – model barely better than random. Indicates features lack separability.

Comparison

Similar for RF/GB; highlights need for better features or balancing.

Section 7: Visualization – Precision-Recall Curve



Explanation

Precision vs Recall. High recall, moderate precision.

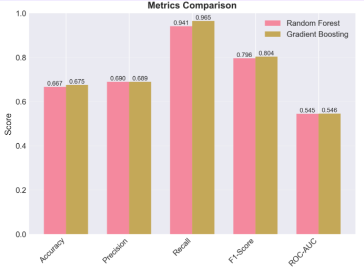
Analysis and Meaning

Good at recalling positives but many false positives. Useful for imbalanced data where recall matters (avoid missing good loans).

Comparison

GB curve slightly better; PR preferred over ROC for imbalance.

Section 7: Visualization – Metrics Comparison



Explanation

Bar chart of Accuracy, Precision, Recall, F1, AUC for RF vs GB.

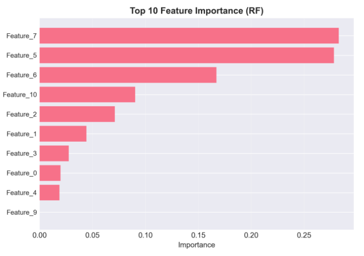
Analysis and Meaning

GB superior in all (e.g., F1 80.39% vs 79.60%) – better balance.

Comparison

Highlights GB's edge in error correction; low AUC common issue.

Section 7: Visualization – Feature Importance



Explanation

Bar chart showing contribution (GB model). Credit$_{H}$*istory* 80%.

Analysis and Meaning

Credit$_{H}$*istorydominates–alignswithQ*$1(80.67\%$*accuracyalone*$)$.*Otherfeaturesaddlittle*; *focuson*

Comparison

Financial > Demographic features; suggests feature selection could simplify model.

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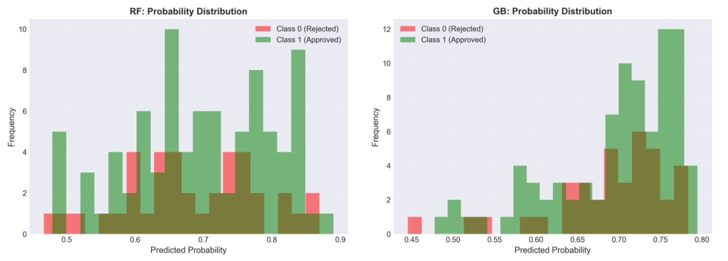
Section 7: Visualization – Probability Distribution



Explanation

Histogram of predicted probabilities for positive class.

Analysis and Meaning

Bias toward high probabilities – overconfidence in approvals, explaining high recall/low precision.

Comparison

Similar for both models; needs calibration (e.g., Platt scaling) for reliable probs.

Conclusion

Summary

Implemented full ML pipeline for loan prediction. Key: Preprocessing cleaned data, PCA reduced dimensions effectively, GB model best with 67.48% accuracy, 96.47% recall, but low 0.5455 AUC.

Key Findings and Analysis

Credit$_{H}$*istorydominantfeature.Modelsbiasedtowardmajorityclass*, *leadingtohighrecallbutmode*

Comparisons

GB > RF > SVM; full pipeline underperforms Q1 single-feature model (67% vs 81%) – suggests over-complexity or preprocessing issues.

Recommendations

Address imbalance (SMOTE), enhance features, calibrate probabilities, try advanced models (XGBoost).

Overall Takeaway

Pipeline demonstrates practical ML but highlights challenges in imbalanced data; improvements needed for deployment.

Questions?

Thank you for your attention

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